

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_failure_replay_compression.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay_compression.
- Best synthetic holdout gap: atsp_no_replay.

Run Metadata

- run_name: run_20260513_172105_i
- started_at_local: 2026-05-13 17:21:05
- finished_at_local: 2026-05-13 17:29:28
- duration_hhmm: 00:08
- duration_seconds: 503.469
- seed_offset: 8000
- replicate_label: i
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.192839	0.009004	0.100922	0.713857	0.58	0.044079
atsp_random_replay	random	score_only	False	0.196068	0.041579	0.118823	0.0	0.78	0.0
atsp_failure_replay	failure	score_only	False	0.19016	0.009004	0.099582	0.869385	0.58	0.007555
atsp_failure_replay_compression	failure	novelty_gate	True	0.162165	0.020176	0.091171	0.786717	0.78	0.026743

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.192839, synthetic 0.009004, combined 0.100922.

- Accepted-epoch count 2, mean accepted code novelty 0.713857, and final complexity 0.58.
- Adaptation efficiency 0.044079 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.192839 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.003915, ry=0.109416.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.196068, synthetic 0.041579, combined 0.118823.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.196068 across 4 instances; family means: ft=0.280377, ftv=0.309361, p=0.019039, ry=0.175496.
- Panel synthetic_holdout mean gap 0.041579 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.080705, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1925, "family": "ftv", "name": "ftv35", "optimality_gap": 0.306857}, {"best_known_cost": 1530, "cost": 1968, "family": "ftv", "name": "ftv38", "optimality_gap": 0.286275}, {"best_known_cost": 1286, "cost": 1492, "family": "ftv", "name": "ftv33", "optimality_gap": 0.160187}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.19016, synthetic 0.009004, combined 0.099582.
- Accepted-epoch count 6, mean accepted code novelty 0.869385, and final complexity 0.58.
- Adaptation efficiency 0.007555 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.19016 across 4 instances; family means: ft=0.319768, ftv=0.33478, p=0.004093, ry=0.101997.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.162165, synthetic 0.020176, combined 0.091171.
- Accepted-epoch count 3, mean accepted code novelty 0.786717, and final complexity 0.78.

- Adaptation efficiency 0.026743 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.162165 across 4 instances; family means: ft=0.27748, ftv=0.318041, p=0.002313, ry=0.050825.
- Panel synthetic_holdout mean gap 0.020176 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.080705, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1797, "family": "ftv", "name": "ftv35", "optimality_gap": 0.219959}, {"best_known_cost": 1530, "cost": 1842, "family": "ftv", "name": "ftv38", "optimality_gap": 0.203922}, {"best_known_cost": 1286, "cost": 1527, "family": "ftv", "name": "ftv33", "optimality_gap": 0.187403}]

Judge Appendix

Summary of Benchmark Comparison

Condition	Replay Mode	Code Novelty (mean)	Synthetic Holdout Gap	Held-Out TSPLIB Gap	Transfer Gap	Compression Pressure	Notes on Transfer vs. Novelty
atsp_no_replay	None	0.714	**0.009004 (best synthetic)**	0.192839	0.100922	No	Highest code novelty but worse transfer gaps than best transfer condition
atsp_random_replay	Random	0.0	0.041579	0.196068	0.118823	No	Zero code novelty; worse transfer and synthetic gaps than failure replay
atsp_failure_replay	Failure	0.869	0.009004	0.19016	0.099582	No	Highest code novelty, slightly better transfer gaps than no replay
atsp_failure_replay_compression	Failure + Compression-aware	0.787	0.020176	**0.162165 (best TSPLIB)**	**0.091171 (best transfer)**	Yes	Transfer metrics improved despite slight code novelty decrease compared to failure_replay

Interpretation

- **Transfer Evidence (Primary Focus):**
- The **best transfer** and **best held-out TSPLIB performance** occur under **atsp_failure_replay_compression** condition, with transfer gap 0.091 and TSPLIB gap 0.162, respectively.
- **No replay (atsp_no_replay)** has the **lowest synthetic gap (0.009)** but significantly worse transfer gaps (TSPLIB 0.193, transfer 0.101).
- **Random replay** performed worst in transfer metrics; also zero code novelty.
- **Code Novelty vs. Transfer:**
- Code novelty is highest in **atsp_failure_replay** (0.869 mean), but transfer gaps are marginally worse than those in compression-aware failure replay.
- Despite some code novelty reduction from failure_replay (0.869) to failure_replay_compression (0.787), transfer metrics improve. This indicates that **transfer gains happen even with reduced code novelty**, emphasizing that

code novelty alone does not guarantee better transfer.

- Random replay with zero code novelty yields the poorest transfer, confirming novelty is necessary but not sufficient.
- **Replay Mode Impact:**
- **No replay** and **failure replay** conditions yield low synthetic holdout gaps (~ 0.009).
- **Random replay** yields poor synthetic and transfer performance, likely due to non-informative samples.
- **Failure replay with compression** leverages replay selectively under compression pressure, achieving best transfer performance.
- **Complexity and Selection Mode:**
- Replay compression conditions yield higher complexity (0.78) versus no replay/failure replay (0.58).
- Selection mode "novelty_gate" in failure replay compression correlates with better transfer despite lower accepted epoch count (3 vs 6 in failure replay).

Conclusion

- **Replay-aware methods, especially failure replay combined with compression-aware selection, provide superior transfer to held-out TSPLIB ATSP instances.**
- This transfer improvement comes **despite a moderate reduction in code novelty**, implying that architectural/selection refinements aid transfer beyond mere novelty.
- **Random replay is ineffective** in this context.
- Synthetic holdout optimality gaps favor no replay and failure replay equally, suggesting synthetic benchmarks alone are insufficient proxy for transfer performance.
- Optimality-gap metrics emphasize the **value of failure replay with compression** as the best overall transfer condition.